

Using the near real-time effective reproduction number R_t as an early-warning tool for seasonal bronchiolitis and influenza-like illness epidemics

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Abstract

Yearly bronchiolitis and influenza-like illness epidemics in France often involve high morbidity and mortality, which severely impact health care. Epidemics are declared by the French National Institute of Public Health based on syndromic surveillance of primary care and emergency departments (EDs), using statistics-based alarms. Although the effective reproduction number (R_t) is used to monitor the dynamics of epidemics, it has never been used as an early-warning tool for bronchiolitis or influenza-like illness epidemics in France. We assessed whether R_t is useful for detecting seasonal epidemics by comparing it to the tool currently used (MASS) by epidemiologists to declare epidemic phases. We used anonymized ED syndromic data from the Île-de-France region in France from 2010 to 2022. We estimated R_t and compared the indication of accelerated transmission ($R_t > 1$) to the MASS epidemic alarm time points. We computed the difference between those 2 time points, time to epidemic peak, and the daily cases documented at first indication and peak. R_t provided alarms for influenza-like illness and bronchiolitis epidemics that were, respectively, a median of 6 days (IQR, 4, 8) and 64 days (IQR, 52, 80) earlier than the alarms provided by MASS. R_t detected earlier signals of bronchiolitis and influenza-like illness epidemics. Using this early-warning indicator in combination with others to declare an annual epidemic could provide opportunities to improve health care system readiness.

Key words: influenza; bronchiolitis; early warning; reproduction number; epidemic.

Introduction

Health surveillance has been defined as the “systematic ongoing collection, collation, and analysis of data and the timely dissemination of information to those who need to know so that action can be taken.”¹ Ever since the earliest attempts to implement health surveillance,² the manner in which information is derived from surveillance data has evolved constantly. Although case definitions are still used to count cases and to compute percentages or ratios (depending on the nature of the denominator), increasing computational power and more developed statistical software have transformed epidemiologists’ capacity to routinely analyze surveillance data. In the context of assessing initial outbreaks of a disease, one of the quantitative sea changes in surveillance in recent years is that modeling—specifically, the computation of the effective reproduction number (R_t)—has become widespread.³

The R_t is defined as the average number of secondary cases infected by a primary case in a population at a point in time during an outbreak in a given area. This time-varying R_t value is an important indicator used to assess and monitor ongoing

epidemics,⁴ anticipate their impact, and assess the effectiveness of interventions.

R_t has never been used in France as an early-warning tool, especially for recurring seasonal epidemics, such as bronchiolitis and influenza-like illnesses (ILIs). Bronchiolitis epidemics are mostly caused by respiratory syncytial virus (RSV),⁵ while seasonal ILI epidemics are closely associated with the circulation of 1 or several influenza viral strains. Syndromic surveillance of bronchiolitis and ILI detects seasonal RSV and influenza epidemics, respectively, with which they are highly correlated.^{6–9} These seasonal epidemics can severely impact health systems, especially hospital capacities, resulting in substantial morbidity and mortality burdens.

Specific preparedness plans are often implemented to ensure hospitals keep functioning during such epidemics. Early-warning tools can help activate prevention strategies and epidemic preparedness by delaying nonurgent admissions, preparing the referral of excess patients to other hospitals, and reallocating human and other resources to priority wards. This reduces the impact

on hospitals, the loss of chance, and unfavorable outcomes for hospitalized patients, especially at the peak of an epidemic.

We used syndromic emergency department (ED) referral data to compare the instantaneous reproduction number R_t with the conventional surveillance method (MASS) used by epidemiologists to declare an epidemic in France. Our aim was to assess whether R_t could detect earlier signals of seasonal bronchiolitis and ILI epidemics onset in the Île-de-France region (IdF—which covers Paris and its surrounding area, amounting to 12.3 million inhabitants).

Methods

OSCOUR surveillance system and data

The OSCOUR network is the main component for ED-based syndromic surveillance by Santé publique France (SpF—the French Public Health Agency).^{10–16} Implemented in 2004, this system automatically collects near-exhaustive digital and anonymized data on clinical diagnoses following ED visits in France, using the 10th revision of the International Classification of Diseases (ICD-10). For epidemiologic surveillance, clinical diagnoses are then grouped into different syndromes of interest, for example, ILI, bronchiolitis, and COVID-19. Data for these syndromes are used routinely to assess the impact of public health events such as heat waves and seasonal epidemics, at both national and subnational levels. In IdF, 98 EDs participate in OSCOUR for a total of over 3.3 million ED visits per year. This amounts to 87.5% of all ED visits in the region.

We extracted the daily number of ED visits in IdF for patients of all ages with clinical diagnoses grouped under the ILI and under the bronchiolitis syndromes in children aged under 2 years, between August 1, 2010, and January 15, 2022. Referrals for ILI were those with ICD-10 codes J09, J10, J11, and their subcodes, as per the World Health Organization ICD-10 classification. Referrals for bronchiolitis were those with ICD-10 codes J21.0, J21.8, and J21.9 in children aged under 2 years, as per the syndromic surveillance case definition.

Data smoothing

Even during an epidemic, the number of ED visits varies according to the day of the week. We therefore used a 2-step approach, as developed in the context of the COVID-19 pandemic. Briefly, this method enabled us to remove the day-of-the-week pattern from patient visits to date by fitting a polynomial regression over the preceding 8 weeks. We then smoothed the incidence by accounting for real time according to the least revision principle.¹⁷ This approach stabilizes R_t estimate trends over time, with a small concession in responsiveness if the data are used on a day-to-day basis.

Estimation of effective reproductive number

The R_t was estimated using the method developed by Cori et al,¹⁸ which requires the distribution of the serial interval (SI). Based on previous studies, we set $\mu_{SI} = 7.5$ days with a $sd_{SI} = 2.1$ days for RSV, represented in our study using bronchiolitis, and $\mu_{SI} = 2.6$ days with a $sd_{SI} = 1.5$ days for influenza, represented here using flu-like illness.^{18,19} A time window for estimating R_t also had to be set; this corresponds to the number of previous days over which transmissibility was averaged. For the present study, we used a period of 7 days. This enabled us to include enough data while keeping sufficient responsiveness to reexamine trends in the R_t estimates over time. The R_t and its 95% credible interval were estimated using the previously smoothed data. Periods of

$R_t > 1$ (ie, indicating an ascending phase of the epidemic), $R_t < 1$ (indicating a descending phase), or an R_t not significantly different from 1 were determined based on whether the 95% credible interval contained the value 1.

Reference epidemic periods using MASS

As a reference to evaluate the relevance of R_t for detecting an epidemic, we used the previously validated MASS tool²⁰ developed by Santé publique France to identify when the number of documented ED visits indicated the onset of a phase of an epidemic (ie, ascending or descending phase). MASS is routinely used each year in France in the surveillance of bronchiolitis and ILI. It combines 3 different detection methods (periodic regression, robust periodic regression, and hidden Markov chains), which take into account the dynamics of ED referrals for the disease concerned, documented over the previous 5 years to define the signal. These statistical outbreak detection models can therefore be altered by a highly unusual dynamic in a given year. Each MASS model is used for a specific geographical area under consideration. MASS model alarms are compared: if 2 of the 3 models produce a signal, this defines a “pre-epidemic phase,” and if all 3 models produce alarms, then this signals an “epidemic phase” (ie, ascending or descending). The final decision to declare an epidemic—and its end—is made by local epidemiologists, who take into account MASS findings and other data sources (intensive care unit surveillance, virological surveillance data, etc).

Inclusion and exclusion criteria

Only seasons that presented at least 1 epidemic alarm, based on the R_t and/or all 3 MASS models, were included in the present analysis. The 2020–2021 season was excluded because of the effects of COVID-19 lockdowns and curfews on ILI and bronchiolitis case counts.

Primary endpoint

We sought to document the time lag (in days) between the first day of the season for which the R_t was significantly greater than 1 (ie, epidemic phase alarm) and the first day of the season when the 3 MASS models all produced alarms. To do this, we used the average of this time lag indicator across all included seasons as the primary endpoint. A positive average time lag meant an $R_t > 1$ alarm occurred earlier than the 3-model MASS alarm. We also computed the time lag between the first day of the alarm and the date of the epidemic peak (ie, the day when the most cases are reported) for either R_t or MASS’s 3 models, respectively. Moreover, we calculated the total number of reported cases and the number of cases at the peak of the epidemic for each syndrome and season and determined the number of cases documented when the respective alarm type was triggered. Finally, we visually examined epidemiologic curves for trend breaks, that is, when a change in outbreak dynamics occurred because the case count increased or decreased more than once during the season.

Sensitivity analysis

To assess whether the 3-model MASS alarm was undermined by any single model, we performed a sensitivity analysis separately comparing each of the 3 models to the R_t estimate. We also compared the R_t alarm with the corresponding 2-model alarm for MASS’s “pre-epidemic phase.”

Statistical analyses

Continuous variables are presented as medians (IQR). Categorical variables are shown with the percentages. All analyses were

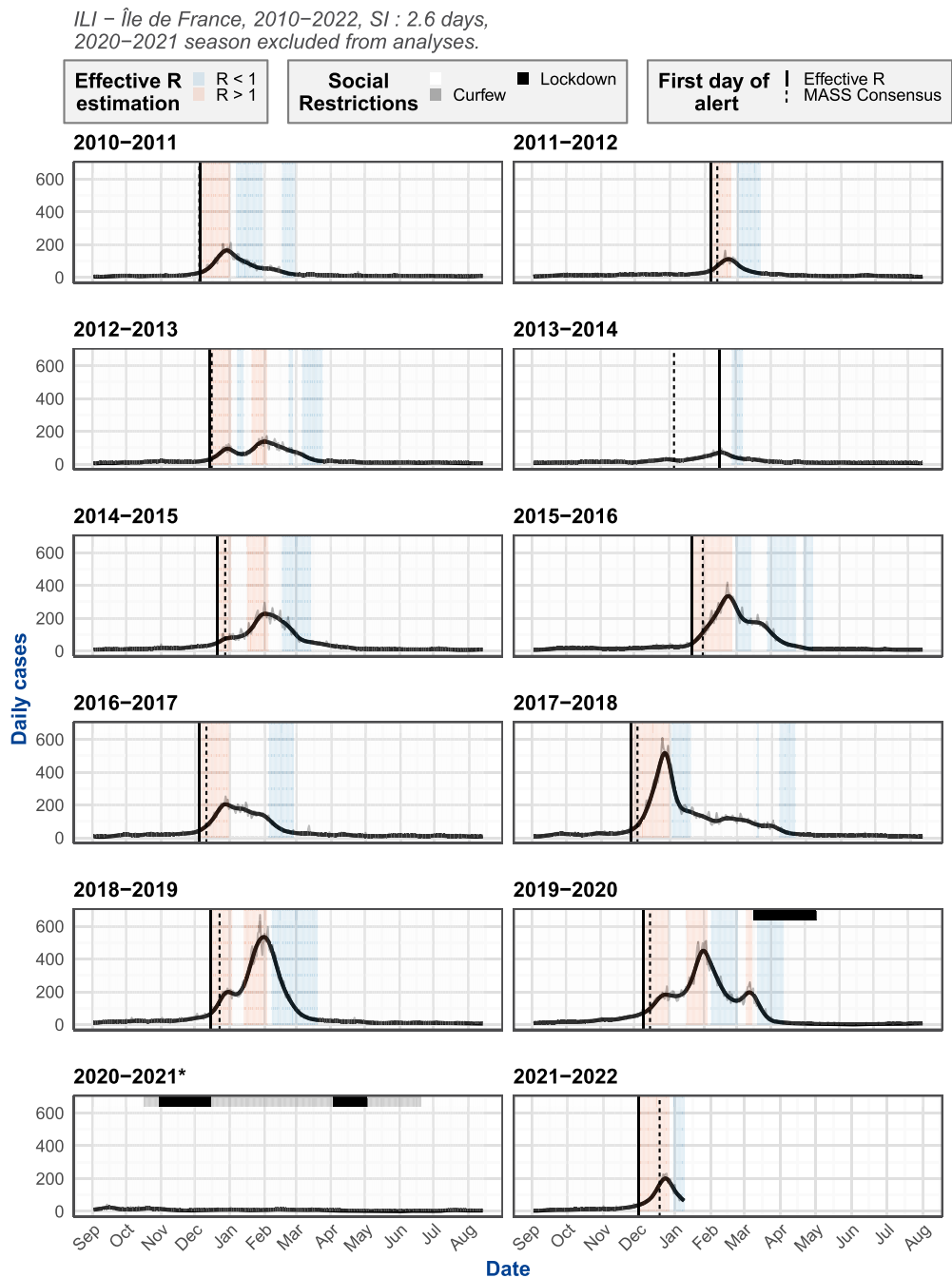


Figure 1. Number of raw and smoothed daily emergency department attendance with R_t estimation significantly different from 1 and the start of MASS epidemic alert, by surveillance season, influenza-like illness, 2010–2022, Île-de-France.

conducted using R 4.1.1 (<https://www.Rproject.org/>) while the EpiEstim (V2.2-4) package was used to estimate R_t . Under French law, our comparison of different computational methods using anonymized data from an approved surveillance system did not require ethical approval.

Results

Epidemic curves for ILI and bronchiolitis are shown in [Figures 1](#) and [2](#), with dates and durations of COVID-19 lockdowns and curfews in 2020–2021. [Supplementary Figures S1](#) and [S2](#) depict the R_t estimations for the corresponding time frame. These curves show the time at which an alarm was triggered by MASS and the

time it would have been triggered using R_t . [Figure 1](#) shows the daily number of cases for each surveillance season, the period during which ILI cases were above the MASS epidemic alarm (ie, all 3 models) threshold, and R_t values significantly different from 1. [Table 1](#) describes the characteristics of each surveillance season, including the dates and the numbers of ED consulting patients needed for the alarm to be triggered by thresholds being crossed in all 3 MASS models (consensus) or by R_t estimations being significantly greater than 1. Dates on which daily case counts peaked are also shown in [Table 1](#).

For example, in 2018–2019, computations were based on a total of 27 290 ED referrals for ILI. The 3-model MASS alarm was triggered on December 23, 2018, with 170 ILI cases, while the R_t

Bronchiolitis – Île de France, 2010–2022, SI : 7.5 days, with the 2020–2021 season excluded from analyses.

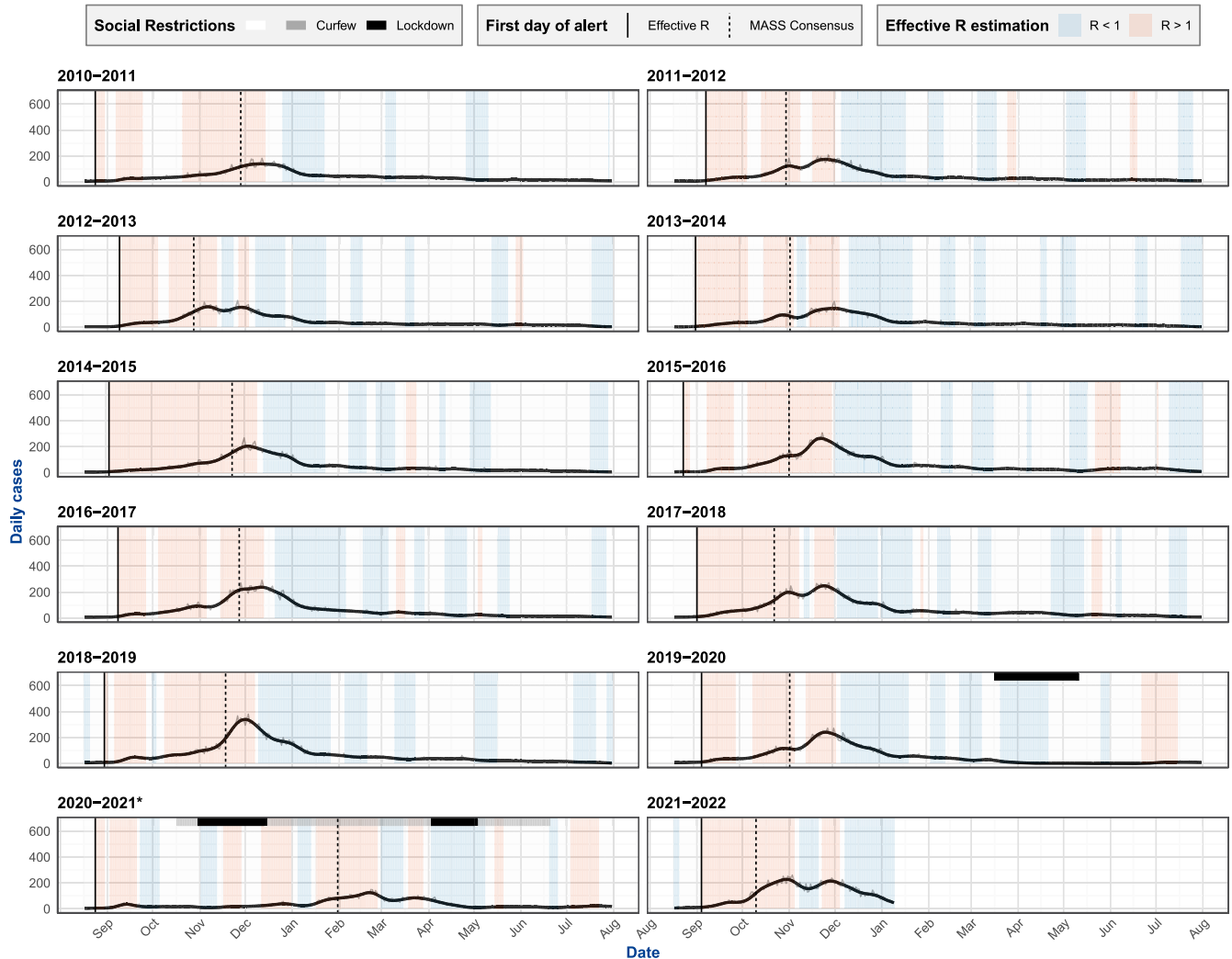


Figure 2. Number of raw and smoothed daily emergency department attendance with R_t estimation significantly different from 1 and the start of MASS consensus alert, by surveillance season for bronchiolitis, 2010–2022, Île-de-France.

alarm would have been triggered 8 days earlier, on December 15, 2018, with 52 ILI cases. The ILI epidemic peaked on February 1, 2019, 40 and 48 days after the MASS and R_t alarms, respectively.

Influenza-like illness

We analyzed 11 ILI epidemics that were documented in IdF during the 12-year study period. Most began between December and January, except for the 2013–2014 season. The median epidemic duration was 11 (9, 12) weeks. Outliers to this pattern were the 2011–2012 (5 weeks) and the 2021–2022 (2 weeks) epidemics. Five epidemics showed a trend break, while the other 6 had only 1 ascending and 1 descending phase.

In 2021–2022, using the R_t would have detected an increase in cases at a time when the number of ED daily visits for ILI was 75% lower than the number required for the MASS alarm (ie, all 3 model alarms) to be triggered (40 vs 170 visits). In terms of early warning, in 9 of the 11 ILI epidemics, the R_t alarm would have been triggered 6 (4, 8) days before the MASS alarms. In terms of the number of days before reaching the peak of the epidemic, the R_t would have triggered an alarm 30 (23, 45) vs 25 (20, 41) days for the MASS alarms.

Of all 11 epidemics, the 2013–2014 outbreak was the latest to start and had the smallest impact on ED; this resulted in ED capacity not being overwhelmed. The R_t was only significantly greater than 1 on 1 day, specifically 2 days before the peak.

Finally, for the 2010–2011 epidemic, the R_t alarm would have been triggered 1 day after the MASS alarm.

Bronchiolitis

As for ILI, we analyzed 11 bronchiolitis epidemics documented in IdF during the 12-year study period. Figure 2 and Table 1 describe the characteristics of each surveillance season. Epidemic curves based on ED visits show a trend break in 7 of the 11 epidemics, marked by the fall holidays in late October and the end-of-year holiday season. The MASS alarm was triggered in either late October or early November for 6 of 11 epidemics, and in most cases, epidemics were over by January (except in 2020–2021). Table 1 shows that the R_t would have triggered a median of 64 (IQR, 52, 80) days before MASS for all 11 epidemics. For 2021–2022, it also shows that up to 94% (8 vs 110) fewer cases would have been needed compared to MASS on the day the R_t alarm was triggered in October 2021.

Table 1. Case counts, dates, and difference in days between the 3-model MASS alarm and the R_t alarm, for influenza-like illness (ILI) and bronchiolitis seasonal epidemics, Île-de-France, 2010-2022.

Influenza-like illness										
Year	Total cases	Consensus ^a		R_t		Difference between consensus— R_t , d	Peak		Time to peak, d	
		Date	Cases	Date	Cases		Date	Cases	Consensus	R_t
2010-2011	9777	Dec 05	50	Dec 06	46	−1	Dec 30	182	25	24
2011-2012	7589	Feb 12	102	Feb 06	40	6	Feb 22	111	10	16
2012-2013	11 320	Dec 16	44	Dec 14	29	2	Feb 02	156	48	50
2013-2014	6930	Jan 05	29	Feb 14	68	−40	Feb 16	97	42	2
2014-2015	15 918	Dec 28	69	Dec 21	55	7	Feb 01	295	35	42
2015-2016	19 379	Jan 31	153	Jan 21	38	10	Feb 23	346	23	33
2016-2017	15 063	Dec 11	90	Dec 05	47	6	Dec 28	252	17	23
2017-2018	26 190	Dec 03	87	Nov 27	71	6	Dec 27	531	24	30
2018-2019	27 290	Dec 23	170	Dec 15	52	8	Feb 01	554	40	48
2019-2020	26 993	Dec 15	90	Dec 09	75	6	Jan 31	496	47	53
2020-2021 ^a	3284						Sep 16	27		
2021-2022	6095	Dec 19	170	Dec 01	40	18	Dec 24	226	5	23
Median (IQR)	15 063 (8683, 22 784)	Dec 19	90 (60, 128)	Dec 14	47 (40, 62)	6 (4, 8)	Feb 01	252 (169, 421)	25 (20, 41)	30 (23, 45)
Bronchiolitis										
Year	Total cases	Consensus		R_t		Difference between consensus— R_t , d	Peak		Time to peak, d	
		Date	Cases	Date	Cases		Date	Cases	Consensus	R_t
2010-2011	14 807	Nov 28	152	Aug 24	9	96	Dec 11	144	13	109
2011-2012	15 834	Oct 30	149	Sep 07	7	53	Nov 25	150	26	79
2012-2013	15 964	Oct 28	113	Sep 09	12	49	Nov 07	162	10	59
2013-2014	14 678	Nov 03	101	Sep 02	5	62	Dec 02	198	29	91
2014-2015	17 240	Nov 23	168	Sep 03	6	81	Dec 03	202	10	91
2015-2016	21 413	Nov 01	134	Aug 23	8	70	Nov 22	257	21	91
2016-2017	20 986	Nov 27	207	Sep 08	13	80	Dec 12	292	15	95
2017-2018	23 071	Oct 22	142	Sep 01	8	51	Nov 24	249	33	84
2018-2019	24 314	Nov 18	210	Aug 30	11	80	Dec 01	296	13	93
2019-2020	17 468	Nov 03	119	Sep 06	15	58	Nov 27	247	24	82
2020-2021 ^a	12 347	Jan 31	80	Aug 24	6	160	Feb 21	125	21	181
2021-2022	16 920	Oct 10	110	Sep 04	8	36	Oct 30	197	20	56
Median (IQR)	17 354 (16 203, 21 306)	Nov 02	146 (123, 164)	Sep 03	8 (8, 12)	64 (52, 80)	Nov 26	224 (171, 255)	18 (13, 23)	88 (80, 92)

^aDate on which case counts crossed the computed threshold for all 3 MASS models. This combined consensus triggered the MASS alarm.

Sensitivity analyses

Table 2 presents the results of the sensitivity analyses. We assessed differences when using consensus for only 2 (instead of all 3) MASS models compared to using the R_t and differences for each separate MASS model compared to the R_t . The 2-model configuration did not alter our results, with R_t keeping its 6-day (−1, 8) lead for ILI and having a 58-day (52, 73) lead for bronchiolitis. For the 3 separate MASS models, the R_t alarm leads over the robust periodic regression alarm algorithm (2 [−1, 6] days) and over the hidden Markov chains alarm (4 [0, 6] days) were similar to those in the 3-model configuration for ILI. With regard to bronchiolitis, no difference in time lags was observed for the 3 separate MASS alarm algorithms compared to the 3-model configuration.

Discussion

This original study evaluates the relevance of using R_t as a new early warning tool for the surveillance of seasonal communicable diseases in France. Using data for IdF (France's most populated

and densely populated region), we found that using R_t would have provided an early-warning alarm for ILI epidemics a median of 6 days (IQR, 4, 8) before the currently used tool (MASS) and 30 days before the epidemic peak. For bronchiolitis, these values were 64 days (IQR, 52, 80) and 88 days, respectively.

There is a large degree of between-season heterogeneity in ILI epidemics in Île-de-France in terms of the start date and level of viral circulation due in part to the introduction of new strains and weather conditions. In contrast, bronchiolitis epidemics are much less heterogeneous. Our results also highlighted differences in the relative earliness of the R_t signal between bronchiolitis (64-day lead over MASS) and ILI (6-day lead). This is partly due to the serial interval for RSV being longer than that for influenza. It is also partly due to the fact that our surveillance data were syndromic, as RSV causes the majority of, but not all, bronchiolitis cases.²¹

In France, the final decision to declare an epidemic is currently made by local epidemiologists, who take into account MASS alarms and other data sources (virological surveillance data, influenza-confirmed intensive care unit admissions,

Table 2. Sensitivity analysis for all years, according to a 2-model (instead of all 3) configuration for the MASS alarm and according to each of the 3 MASS models separately for influenza-like illness (ILI) and bronchiolitis seasonal epidemics, Île-de-France, 2010-2022.

Influenza-like illness																
Year	Two-model MASS alarm				Periodic regression (PR) MASS model				Robust periodic regression (RPR) MASS model				Hidden Markov chains (HMC) MASS model			
	Date	Cases	2 MASS models vs R_t , d		Date	Cases	PR vs R_t , d		Date	Cases	RPR vs R_t , d		Date	Cases	HMC vs R_t , d	HMC vs peak, d
			R_t , d	peak, d			PR vs peak, d	RPR vs peak, d								
2010-2011	05 Dec	50	-1	25	05 Dec	50	-1	25	05 Dec	50	-1	25	05 Dec	50	-1	25
2011-2012	05 Feb	44	-1	17	12 Feb	102	6	10	05 Feb	44	-1	17	29 Jan	36	-8	24
2012-2013	04 Nov	31	-40	90	04 Nov	31	-40	90	04 Nov	31	-40	90	16 Dec	44	2	48
2013-2014	05 Jan	29	-40	42	05 Jan	29	-40	42	05 Jan	29	-40	42	22 Dec	23	-54	56
2014-2015	28 Dec	69	7	35	28 Dec	69	7	35	28 Dec	69	7	35	21 Dec	55	0	42
2015-2016	31 Jan	153	10	23	31 Jan	153	10	23	31 Jan	153	10	23	24 Jan	68	3	30
2016-2017	11 Dec	90	6	17	11 Dec	90	6	17	11 Dec	90	6	17	11 Dec	90	6	17
2017-2018	03 Dec	87	6	24	03 Dec	87	6	24	03 Dec	87	6	24	03 Dec	87	6	24
2018-2019	23 Dec	170	8	40	23 Dec	170	8	40	16 Dec	86	1	47	23 Dec	170	8	40
2019-2020	08 Dec	89	-1	54	08 Dec	89	-1	54	24 Nov	34	-15	68	15 Dec	90	6	47
2020-2021 ^a																
2021-2022	12 Dec	83	11	12	19 Dec	170	18	5	05 Dec	40	4	19	12 Dec	83	11	12
Median (IQR)	11 Dec	85 (55, 90)	6 (-1, 8)	24 (18, 39)	15 Dec	90 (74, 140)	6 (1, 8)	24 (18, 39)	08 Dec	60 (41, 87)	2 (-1, 6)	24 (20, 44)	15 Dec	76 (51, 89)	4 (0, 6)	28 (24, 42)
Bronchiolitis																
Year	2 MASS models				1 MASS model: PR				1 MASS model: RPR				1 MASS model: HMC			
	Date	Cases	2 MASS models vs R_t , d		Date	Cases	PR vs R_t , d		Date	Cases	RPR vs R_t , d		Date	Cases	HMC vs R_t , d	HMC vs peak, d
			R_t , d	peak, d			PR vs peak, d	RPR vs peak, d								
2010-2011	28 Nov	152	96	13	28 Nov	152	96	13	28 Nov	152	96	13	14 Nov	72	82	27
2011-2012	30 Oct	149	53	26	30 Oct	149	53	26	30 Oct	149	53	26	23 Oct	91	46	33
2012-2013	21 Oct	98	42	17	21 Oct	98	42	17	28 Oct	113	49	10	21 Oct	98	42	17
2013-2014	27 Oct	105	55	36	27 Oct	105	55	36	03 Nov	101	62	29	27 Oct	105	55	36
2014-2015	16 Nov	141	74	17	16 Nov	141	74	17	23 Nov	168	81	10	02 Nov	97	60	31
2015-2016	25 Oct	116	63	28	25 Oct	116	63	28	01 Nov	134	70	21	25 Oct	116	63	28
2016-2017	20 Nov	154	73	22	20 Nov	154	73	22	27 Nov	207	80	15	30 Oct	104	52	43
2017-2018	22 Oct	142	51	33	15 Oct	112	44	40	22 Oct	142	51	33	22 Oct	142	51	33
2018-2019	11 Nov	117	73	20	11 Nov	117	73	20	18 Nov	210	80	13	04 Nov	77	66	27
2019-2020	03 Nov	119	58	24	03 Nov	119	58	24	03 Nov	119	58	24	27 Oct	123	51	31
2020-2021 ^a	31 Jan	80	160	21	31 Jan	80	160	21	31 Jan	80	160	21	24 Jan	67	153	28
2021-2022	03 Oct	57	29	27	03 Oct	57	29	27	03 Oct	57	29	27	10 Oct	110	36	20
Median (IQR)	30 Oct	119 (110, 146)	58 (52, 73)	24 (18, 28)	30 Oct	117 (108, 145)	58 (48, 73)	24 (18, 28)	03 Nov	142 (116, 160)	62 (52, 80)	21 (13, 26)	27 Oct	104 (94, 113)	52 (48, 62)	31 (27, 33)

clusters in long-term care facilities, etc) before reaching a conclusion. Using R_t in addition to existing data sources and MASS signals to analyze the same surveillance data on ED visits could provide even earlier forewarning for epidemics. As our results indicate, the degree of this added forewarning would vary according to the specific disease. In any case, an improved early warning system could ensure that adequate measures are taken to avoid hospital capacity being overwhelmed, especially before the end-of-year holiday season in the case of seasonal ILI.

Reproduction number

The reproduction number is generally used to determine the intrinsic epidemiologic features of a pathogen—or of a variant—in a given place at a given time, usually in an emerging disease setting.^{4,22,23} In some cases, the characterization of these features has been further refined by comparing R_t to that of a well-established pathogen. One example of this is a comparative study of R_t computed for both influenza and COVID-19 during the ascending phases of concurrent epidemics in Hong Kong.²⁴

However, R_t is complex³ in that it is not a fixed property of a pathogen dependent only on its virulence. As demonstrated by the COVID-19 pandemic, the R_t changes rapidly when (1) a pandemic progresses, (2) infection-induced immunity increases, (3) personal protection or distancing measures are introduced, and (4) widespread testing and vaccination become increasingly accessible. Accordingly, its usefulness in terms of public health is limited beyond a given time, place, and population, and using it for comparisons across different epidemiologic settings is hazardous.²⁵ These various aspects are not an issue, however, if the data provided by a health care structure that is easily accessed by a population—for example, EDs—are used to compute R_t to provide early warning of the onset of a seasonal epidemic and to monitor its progression, as was the case in our study. In the case of ILI and bronchiolitis, R_t as an early-warning tool simply reflects the speed at which the number of cases progresses in the community at a given time, irrespective of whether or not the population concerned observes social distancing or personal protective measures.

Interpretation and advantages of R_t estimates

Epidemic alarms using MASS and R_t estimations provide different information. The former compares the number of new cases to an expected number based on historical data series and triggers an epidemic alarm, irrespective of the pathogen's circulation dynamics in the community. Accordingly, MASS reflects the epidemic's magnitude. In contrast, the R_t provides information as to the dynamic of circulation. It is not used as a prospective tool to anticipate impact. Instead, it is used to indicate whether community transmission is gaining speed over a short time period. It is a measure of transmission growth or decay. Therefore, both measures (ie, MASS and R_t) are complementary. The late and short 2013-2014 influenza season illustrates the relevance of using R_t as a more specific tool to detect an epidemic compared to MASS detection alarms during an ILI epidemic with no major impact on health care systems.

Bias and limitations

Our study has limitations and biases. First, EDs do not systematically use rapid tests to diagnose influenza in IdF. Syndromic surveillance is therefore based on clinical diagnosis, not virological data. Similarly, bronchiolitis syndromic data do not take virological findings into account.

Second, our study sought to compare one biostatistical approach with another, with both being based on the same syndromic case counts. While the quality of OSCOUR data is good, it is not perfect; clinical diagnosis data are not collected by the electronic system for approximately 25% of visits. However, this surveillance bias is nonsystematic, and the sheer amount of ED visits in the IdF region provides ample robustness when estimating trends.

Third, our study only focused on the IdF region of France. This region covers 12.3 million inhabitants in Paris and the surrounding areas and has a high population density. The performance of R_t as an early-warning tool may be lower in less populated, less densely populated areas or areas where access to ED is more difficult.

Fourth, using R_t as an early-warning tool for epidemics may be sensitive to sudden changes in health care-seeking behavior and increases in ED visits. Such changes are infrequent, however, and are usually limited in terms of numbers and duration. They occur during short holidays when community doctors are away, for example; foreseeing an impact on health care structures in these situations was exactly the focus of our study.

Fifth, R_t estimates depend on hypotheses related to the pathogen, including its serial interval. However, the estimates we used were found in the literature and did not vary significantly for influenza subtypes. Furthermore, although these estimates may affect the magnitude of the R_t estimates, they do not affect these estimates' dynamics across time.

Finally, our data-smoothing method reduced the detection of sudden changes in incidence. We chose, however, to smooth the data over a short period of time to reduce noise.

Timing in early-warning systems is essential. On the one hand, the alarm must not come too late or there will be too little time to take action. On the other hand, it must not come too early, as it may conflict with health care providers' priorities and immediate concerns, resulting in no impactful action being taken. With respect to ILI epidemics, the 6-day lead of the R_t over the MASS alarm would be hugely beneficial in that measures to alleviate the epidemic's impact on hospital structures could be put in place immediately, especially before the end-of-year holiday season when health care worker forces are partially depleted. Furthermore, the 25-day time lag we found for R_t before the epidemic peak would ensure that messages could still be disseminated in time to encourage influenza vaccination in people who have not yet received it.

In contrast, in the case of bronchiolitis, a 64-day forewarning is too long to have an immediate impact on hospital preparedness. It is, however, early enough for prevention messages in kindergartens and the community rather than hospitals. The latter can be targeted later by a more focused alert declared by epidemiologists based on complementary data sources. Should COVID-19 infection become the "new normal" and be added to the existing list of seasonal epidemics,^{26,27} using R_t and contextualizing findings may help clinicians and decision-makers gain substantial operational forewarning to mitigate the impact of seasonal epidemics on hospitals and patients.

Conclusion

R_t is a useful early-warning indicator in routine surveillance of ILI and bronchiolitis in the IdF region of France. Early warning allows ED to prepare for emerging seasonal epidemics and enables public health stakeholders to introduce timely control measures and prevention communication campaigns. R_t provided added

value for the early detection of a seasonal epidemic. However, this value differs depending on the disease. R_t must be used in addition to other epidemiologic indicators when epidemiologists decide whether or not to declare an epidemic.

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Supplementary material

Supplementary material is available at the American Journal of Epidemiology online.

Conflict of interest

The authors declare that they have no competing interests.

Data availability

Secondary analysis of data subject to restrictions.

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